

Last Time: Nondeterministic Finite Automata (NFAs)

Today: NFAs are no more computationally powerful than DFAs.

Theorem: Let N be an NFA. Then $L(N)$ is regular.

(Since all DFAs are basically NFAs, $\Rightarrow$ DFAs and NFAs have equivalent computational power.)

Proof strategy: Given an NFA, convert in to a DFA.

Fun With $\hat{\Delta}$

Recall: $\Delta : Q \times \Sigma \rightarrow 2^Q$.

Intuition: $\Delta(q, a)$ = “What states can we get to from q after reading a ?”

$\hat{\Delta}(A, x)$ = “What states can we get to from some state in A after reading x ?”

- $\hat{\Delta}(A, \varepsilon) = A$
- $\hat{\Delta}(A, xa) = \bigcup_{q \in \hat{\Delta}(A, x)} \Delta(q, a)$ for $x \in \Sigma^*$, $a \in \Sigma$.

Exercise for today: Compute $\hat{\Delta}(A, xa)$ for an example.

Example: At least 1 a .

- $\hat{\Delta}(\{1\}, a) = \hat{\Delta}(\{1\}, a) = \{2\}$
- $\hat{\Delta}(\{1\}, b) = \hat{\Delta}(\{1\}, b) = \{1, 2\}$
- $\hat{\Delta}(\{1\}, aa) = \hat{\Delta}(\{2\}, a) = \emptyset$
- $\hat{\Delta}(\{1\}, ab) = \hat{\Delta}(\{2\}, b) = \{2\}$
- $\hat{\Delta}(\{1\}, ba) = \hat{\Delta}(\{1, 2\}, a) = \{2\}$
- $\hat{\Delta}(\{1\}, bb) = \hat{\Delta}(\{1, 2\}, b) = \{1, 2\}$
- $\hat{\Delta}(\{1\}, aaa) = \hat{\Delta}(\emptyset, a) = \emptyset$
- $\hat{\Delta}(\{1\}, aab) = \hat{\Delta}(\emptyset, b) = \emptyset$

Set up subset DFA.

To evaluate $\hat{\Delta}(A, x)$: run DFA on x .

Conclusion: This DFA simulates the NFA!

Given string x , how could I see if NFA accepts it?

Simulate NFA starting in start states, see if I end up in a set containing a final state.

We’ve constructed a DFA capable of simulating an NFA!

The Subset DFA

Given an NFA $N = (Q_N, \Sigma, \Delta_N, S_N, F_N)$.

Goal: Construct a DFA M such that $L(N) = L(M)$

$$M = (Q_M, \Sigma, \delta_M, s_M, F_M)$$

- $Q_M = 2^{Q_N}$ - set of all subsets of states of N .
(How many states for an NFA with k states?)
- Same alphabet Σ .
- $\delta_M(A, a) = \hat{\Delta}_N(A, a)$

$s_M = S_N$ (Start simulating from the set of start states.)

$F_M = \{A \in Q_M \mid A \cap F_N \neq \emptyset\}$ (See if I end up in a set containing a final state.)

Define $\hat{\delta}_M(A, x)$:

- $\hat{\delta}_M(A, \varepsilon) = A$
- $\hat{\delta}_M(A, xa) = \delta_M(\hat{\delta}_M(A, x), a)$

Lemma: $\hat{\Delta}_N(A, x) = \hat{\delta}_M(A, x)$.

Proof: (Structural Induction)

Base case: $\hat{\delta}_M(A, \varepsilon) = A = \hat{\Delta}_N(A, \varepsilon)$ (Definitions of $\hat{\delta}_M, \hat{\Delta}_N$.)

Inductive case: Assume $x = ya$ for $y \in \Sigma^*, a \in \Sigma$.

$$\begin{aligned} \hat{\delta}_M(A, ya) &= \delta_M(\hat{\delta}_M(A, y), a) \text{ (Def. of } \hat{\delta}_M) \\ &= \delta_M(\hat{\Delta}_N(A, y), a) \text{ (Inductive hypothesis)} \\ &= \hat{\Delta}_N(\hat{\Delta}_N(A, y), a) \text{ (Def. of } \delta_M) \\ &= \bigcup_{q \in \hat{\Delta}_N(\hat{\Delta}_N(A, y), \varepsilon)} \Delta_N(q, a) \text{ (Def. of } \hat{\Delta}) \\ &= \bigcup_{q \in \hat{\Delta}_N(A, y)} \Delta_N(q, a) \text{ (Def. of } \hat{\Delta}) \\ &= \hat{\Delta}_N(A, ya) \text{ (Def. of } \hat{\Delta}) \quad \square \end{aligned}$$

Theorem: $L(M) = L(N)$

$$\begin{aligned} x \in L(M) &\Leftrightarrow \hat{\delta}_M(S, x) \in F_M \\ &\Leftrightarrow \hat{\delta}_M(S, x) \cap F_N \neq \emptyset \\ &\Leftrightarrow \hat{\Delta}_N(S, x) \cap F_N \neq \emptyset \\ &\Leftrightarrow x \in L(N) \end{aligned}$$